Open source practice

Specification of your final project demonstration

Demo scenario overview.

Logging in -

- Specific features:

  Users input their data to access an account.

- User actions:

  Users that want to post are redirected to the login page, then enter their email or username and password, and submit.

- Functional parts of the application:

  Login page presented to the user, and data submitted will be saved into a database.

Signing up -

- Specific features:

  Users will input their data to create an account.

- User actions:

  Users that want to login but do not have an account may select the sign-up option, then input their email, username and password to create one.

- Functional parts of the application:

  Sign-up page presented to the user, and data submitted will be saved into a database.

Account recovery -

- <u>Specific features:</u>

  Users will input their email to recover their account information.

- <u>User actions:</u>

  Users that have forgotten their password may click a “forgotten password” button and be redirected to a recovery page to fill in their email. An OTP will be sent to verify the email's rightful owner, then redirected to change password. Then redirected back to the login page.

- <u>Functional parts of the application:</u>

  Select input email to send OTP verification, update corresponding account password in database.

Accessing a forum -

- <u>Specific features:</u>

  Users may look for a forum to browse.

- <u>User actions:</u>

  Users look for a forum and click to get into, there they can see posts and comments.

- <u>Functional parts of the application:</u>

  Posts and their comments from forums that have been chosen will be pulled from the database and displayed.

Creating a post -

- <u>Specific features:</u>

  Users can upload a post.

- <u>User actions:</u>

Users click on a (+) button while in a forum and input the text content they would like to post in the forum.

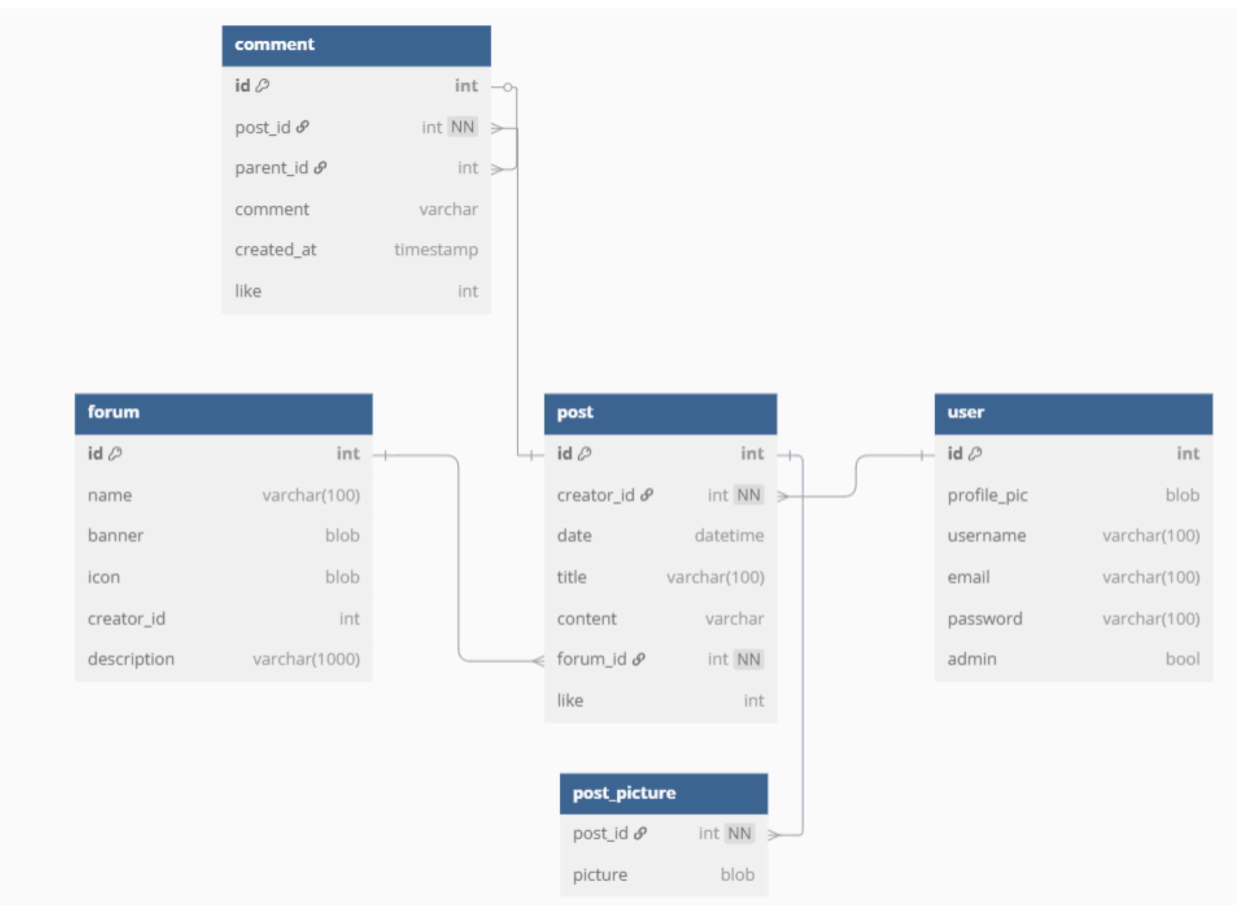
- Functional parts of the application:

  Content written within the post will be read and stored in the database for the relevant forum.

Planned URL Endpoints

URL	Method	expected variable(from query string / form)	Session Variable	Table Operation
forum.php	GET	forum_id, post_id	user_id	SELECT all attributes of a chosen forum from forum_t
homepage. php	GET	forum_id	user_id	SELECT some post from post_t for each forum, SELECT specific attribute from user_t
account.php	GET	user_id	user_id	SELECT user_t attribute values
login.php	POST	Username or email and password		SELECT user_t attributes value
signup.php	POST	Username, email and password		INSERT user_t attribute values
post.php	GET	post_id	user_id	fetch a specific post from post_t, fetch a specific post from post_t
write.php	POST	Title, Content, Image (Nullable)	user_id	INSERT post_t attribute values

Database design



Obeying third normal form

Comment table - All attributes within the comment table (id, post_id, parent_id, comment, created_at, like) are solely dependent on the primary key that is **id**. With post_id and parent_id as foreign keys.

Forum table - The primary key of this table is **id** with creator_id as foreign key to user_id. Meanwhile all attributes depend solely on the table's **id.**

User table - All attributes (profile_pic, username, email, password, admin) depend only on id. And there are no repeating groups.

Post table - Creator_id is a foreign key from the user *table* and forum_id is a foreign key from the forum *table.* The remaining attributes within this table are fully dependent on **id.**

Post picture - pictures in this table fully rely on *post_id*.

Overall, all attributes in the tables in our database obey the third normal form by not having repeating groups within a table, no multivalued attributes and no partial/transitive dependencies.